AP Calculus BC Unit 6 Review Book

Integration and Accumulation of Change

How to use this review book
This guide is detailed, beginner-friendly, and designed for a serious final review. Every major Unit 6 idea is included.
An integral can mean signed area, accumulated change, or the difference of an antiderivative. You must know which meaning the question is using.

Unit 6 Big Picture

Unit 6 connects a rate function, a definite integral, an accumulation function, and an antiderivative. A strong student can move among all four without confusion.

Core formulas

$$\int_a^b f(x)\,dx$$

$$G(x) = \int_a^x f(t)\,dt$$

$$F'(x) = f(x)$$

6.1 Exploring Accumulations of Change

If r(t) is a rate, then integrating r(t) over an interval gives the net accumulated change.

In motion, displacement and total distance are not the same when velocity changes sign.

Core formulas

$$\Delta A = \int_a^b r(t)\,dt$$

$$\text{displacement} = \int_a^b v(t)\,dt$$

$$\text{distance} = \int_a^b |v(t)|\,dt$$

Worked examples

Example: displacement versus distance

Let v(t)=t-2 on [0,4]. Velocity is negative before t=2 and positive after t=2.

$$\int_0^4 (t-2)\,dt = \left[\frac{t^2}{2} - 2t\right]_0^4 = 0$$

$$\int_0^2 (2-t)\,dt + \int_2^4 (t-2)\,dt = 4$$

What AP often asks

- Interpret a definite integral in words and units.
- Explain net change versus total amount.

6.2 Approximating Areas with Riemann Sums

A Riemann sum approximates an integral with rectangles. The subinterval width is delta x, and the rectangle height comes from a chosen sample point.

Core formulas

$$\Delta x = \frac{b-a}{n}$$

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

Worked examples

Example: right Riemann sum for x^2 on [0,4]

Use 4 equal subintervals.

$$R_4 = (1^2 + 2^2 + 3^2 + 4^2)(1) = 30$$

$$\int_0^4 x^2\,dx = \left[\frac{x^3}{3}\right]_0^4 = \frac{64}{3}$$

What AP often asks

- Compute left, right, and midpoint sums.
- Know when an estimate is an overestimate or underestimate.

6.3 Riemann Sums, Summation Notation, and Definite Integrals

A definite integral is the limit of Riemann sums as the number of rectangles goes to infinity.

Core formulas

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

$$\Delta x = \frac{b-a}{n}$$

Worked examples

Example: convert a sigma expression to an integral
The width is 3/n, so the interval length is 3. The sample point is 3i/n, so the interval starts at 0.

$$\lim_{n\to\infty}\sum_{i=1}^{n}\sqrt{1+\frac{3i}{n}}\cdot\frac{3}{n}$$

$$\int_0^3 \sqrt{1+x}\,dx$$

6.4 The Fundamental Theorem of Calculus and Accumulation Functions

FTC Part 1 says that differentiating an accumulation function returns the original integrand.

If the upper limit is a function of x, you must also use the chain rule.

Core formulas

$$G(x) = \int_a^x f(t)\,dt$$

$$G'(x) = f(x)$$

$$\frac{d}{dx}\int_a^{u(x)} f(t)\,dt = f(u(x))u'(x)$$

Worked examples

Example 1
Differentiate the accumulation function.

$$G(x) = \int_2^x \sin(t^2)\,dt$$

$$G'(x) = \sin(x^2)$$

Example 2
Now the upper limit is x^3, so chain rule is required.

$$H(x) = \int_0^{x^3} \frac{1}{1+t^2}\,dt$$

$$H'(x) = \frac{3x^2}{1+x^6}$$

6.5 Interpreting the Behavior of Accumulation Functions

If G(x)=integral from a to x of f(t) dt, then the sign of f controls whether G increases or decreases. The derivative of f controls the concavity of G.

Core formulas

$$G'(x) = f(x)$$

$$G''(x) = f'(x)$$

What AP often asks

- G increases where f is positive.
- G decreases where f is negative.
- G has a local maximum when f changes from positive to negative.
- G has a local minimum when f changes from negative to positive.
- G is concave up where f' is positive and concave down where f' is negative.

6.6 Applying Properties of Definite Integrals

Many AP questions in this topic are pure algebra. Memorize the standard properties.

Core formulas

$$\int_a^a f(x)\,dx = 0$$

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$$

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$$

$$\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

Worked examples

Example: combine known integrals
If the integral from 1 to 4 is 5 and the integral from 4 to 7 is -2, add them.

$$\int_1^7 f(x)\,dx = 3$$

6.7 The Fundamental Theorem of Calculus and Definite Integrals

FTC Part 2 is the evaluation rule: find an antiderivative and subtract lower from upper.

Core formulas

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Worked examples

Example 1
Polynomial integrand.

$$\int_1^3 2x\,dx = [x^2]_1^3 = 8$$

Example 2
Trigonometric integrand.

$$\int_0^\pi \sin x\,dx = [-\cos x]_0^\pi = 2$$

6.8 Finding Antiderivatives and Indefinite Integrals

An indefinite integral gives all antiderivatives, so every answer must include +C.

Core formulas

$$\int f(x)\,dx = F(x) + C$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ \ n \neq -1$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int \sec^2 x\,dx = \tan x + C$$

Worked examples

Example
Integrate term by term.

$$\int \left(3x^2 + \frac{1}{x} - 4\cos x\right)dx = x^3 + \ln|x| - 4\sin x + C$$

6.9 Integrating Using Substitution

Substitution is the reverse of the chain rule. It is most useful when the integrand contains an inside function and its derivative.

Core formulas

$$u = g(x),\ \ du = g'(x)\,dx$$

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

Worked examples

Example 1
Let u=x^2.

$$\int 2x\cos(x^2)\,dx = \int \cos u\,du = \sin(x^2) + C$$

Example 2

In a definite integral, changing the limits is often the cleanest approach.

$$\int_0^2 \frac{2x}{x^2+1}\,dx = \int_1^5 \frac{1}{u}\,du = \ln 5$$

6.10 Long Division and Completing the Square

For top-heavy rational functions, do polynomial long division first. For quadratic denominators, complete the square and look for an arctangent form.

Core formulas

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

Worked examples

Example 1: long division

Rewrite before integrating.

$$\int \frac{x^2+1}{x}\,dx = \frac{x^2}{2} + \ln|x| + C$$

Example 2: complete the square

Convert the denominator to a shifted square plus a constant.

$$x^2 + 4x + 13 = (x+2)^2 + 9$$

$$\int \frac{1}{x^2+4x+13}\,dx = \frac{1}{3}\arctan\left(\frac{x+2}{3}\right) + C$$

6.11 Integration by Parts (BC Only)

Integration by parts comes from the product rule and is useful for products that do not simplify nicely under substitution.

Core formulas

$$\int u\,dv = uv - \int v\,du$$

Worked examples

Example 1

Choose u=x and dv=e^x dx.

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C$$

Example 2

Rewrite the integral of ln(x) as ln(x) times 1.

$$\int \ln x\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$$

6.12 Linear Partial Fractions (BC Only)

Use partial fractions for proper rational functions with factorable denominators.

Worked examples

Example

Decompose first, then integrate the simpler pieces.

$$\frac{1}{x^2-1} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$\int \frac{1}{x^2-1}\,dx = \frac{1}{2}\ln|x-1| - \frac{1}{2}\ln|x+1| + C$$

6.13 Improper Integrals (BC Only)

An integral is improper if the interval is infinite or the integrand is unbounded. Rewrite it as a limit and then decide whether that limit exists.

Worked examples

Example 1: convergent

Infinite interval with a p-type integrand.

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\int_1^b x^{-2}\,dx$$

$$\lim_{b\to\infty}\left[-\frac{1}{x}\right]_1^b = 1$$

Example 2: divergent

This one grows without bound.

$$\int_1^{\infty} \frac{1}{x}\,dx = \lim_{b\to\infty}[\ln x]_1^b = \infty$$

Example 3: unbounded endpoint
Use a one-sided limit.

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{a\to 0^+}\int_a^1 x^{-1/2}\,dx$$

$$\lim_{a\to 0^+}[2\sqrt{x}]_a^1 = 2$$

6.14 Selecting Techniques for Antidifferentiation

Method selection is a major Unit 6 skill. Classify the integrand before you start manipulating it.

What AP often asks

- Basic antiderivative when the answer is immediate.
- Substitution when an inside function and its derivative both appear.
- Long division for top-heavy rational functions.
- Completing the square for a quadratic denominator leading to arctangent form.
- Integration by parts for products that substitution does not simplify.
- Partial fractions for proper rational functions with factorable denominators.
- Improper-integral rewriting when the interval is infinite or the integrand is unbounded.

High-Frequency Formula Sheet

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\,\Delta x$$

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$$

$$\frac{d}{dx}\int_a^{u(x)} f(t)\,dt = f(u(x))u'(x)$$

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C,\ \ n \neq -1$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

$$\int u\,dv = uv - \int v\,du$$

Common Mistakes Checklist

- Forgetting that total distance uses absolute value of velocity.
- Forgetting the chain rule when applying FTC Part 1 to a nontrivial upper limit.
- Using the power rule incorrectly on the integral of 1/x.
- Forgetting to change limits in a definite-integral substitution problem.
- Forgetting +C in an indefinite integral.
- Adding +C to a definite integral answer.
- Using integration by parts when substitution is simpler.
- Not checking convergence of an improper integral.